Pimpri-Chinchwad Education Trust's
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Assignment No 3

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Subject: Design and Analysis of Algorithms
Batch: A3
GitHub Link: https://github.com/SOHAM0007-CODER/DAA_ASSIGNMNET

1. Topic

A flood-relief boat has a limited carrying capacity W. Each supply item has a weight and utility value. Use Fractional Knapsack to select whole items or portions of divisible supplies so that total utility is maximized without exceeding the boat capacity.

2. Course Objective

- Understand the computational complexity of algorithms.
- Select a suitable algorithm design strategy for a practical problem.

3. Course Outcome

- Apply a Greedy strategy based on utility per unit weight.
- Compute fractional selections and analyze the conditions for greedy optimality.

Theory

Fractional Knapsack allows part of an item to be taken. Compute each item's profit-to-weight ratio, sort ratios from highest to lowest, and select supplies in that order. If the next item does not fully fit, take the fraction that fills the remaining capacity.

Each input has a supply name, weight and utility. All rows represent divisible bulk lots, not indivisible physical pieces: a fraction of the blanket lot means some blankets, not a cut blanket. The sample takes medicine in full. A mixture requiring indivisible packages needs a different model.

Working of the Algorithm

- 1. Read named supply lots, carrying capacity, weights and utilities.
- 2. Print the divisibility and utility-per-kg table.
- 3. Sort ratios descending and load whole lots or a final partial lot.
- 4. Print every decision, percentage, loaded weight, utility and boat contents.

Complexity Analysis

Computing ratios takes O(N), sorting takes O(N log N), and the selection pass takes O(N). Therefore, the overall sorting-based method takes O(N log N) time.

Time Complexity

Best Case: O(N log N) with the sorting-based implementation.

Average Case: O(N log N).

Worst Case: O(N log N); sorting dominates the linear selection pass.

Space Complexity

O(N) total storage for the item records. Selection uses constant extra scalar variables; a typical comparison sort also uses O(log N) stack space.

Algorithm - Steps

- Step 1: Calculate ratios and arrange items from highest ratio to lowest.
- Step 2: Fill the boat with whole items while they fit.
- Step 3: Use a fraction for the final item if needed and add its proportional utility.

Pseudocode

```
sort items by decreasing profit/weight
remaining = capacity; totalProfit = 0
for each item in sorted order:
    if remaining == 0: break
    fraction = min(1, remaining / item.weight)
    totalProfit += fraction * item.profit
    remaining -= fraction * item.weight
print fractions, weight used and totalProfit
```

Flow of Execution

START -> Read supplies -> Calculate ratios -> Sort -> Select whole items or fractions -> Print result -> END.

6. Example

Capacity is 60 kg. Blankets: (20,60), Food: (20,100), BabyFood: (5,30), Medicine: (30,120), Water: (15,40). Load BabyFood, Food and Medicine fully (55 kg), then 25% of the blanket lot (5 kg). Water is left behind. Maximum utility is 265.

Conclusion

Named bulk supplies make the fractional selections explicit. Greedy by utility/weight is optimal for divisible lots with positive weights and nonnegative utilities. Whole-package requirements belong to 0/1 Knapsack.

Assignment 3 | SAMPLE INPUT (input.txt)

```
5 60
Blankets 20 60
Food 20 100
BabyFood 5 30
Medicine 30 120
Water 15 40
```

Assignment 3 | FULL PROGRAM OUTPUT

```
ASSIGNMENT 3: FLOOD RELIEF BOAT - FRACTIONAL KNAPSACK
Enter item count and boat capacity (kg):
Enter each bulk lot: name weight utility (one-word names)

SUPPLY AND DIVISIBILITY TABLE
Supply | Weight kg | Utility | Utility/kg | Divisibility
Blankets | 20.00 | 60.00 | 3.00 | Divisible bulk lot
Food | 20.00 | 100.00 | 5.00 | Divisible bulk lot
BabyFood | 5.00 | 30.00 | 6.00 | Divisible bulk lot
Medicine | 30.00 | 120.00 | 4.00 | Divisible bulk lot
Water | 15.00 | 40.00 | 2.67 | Divisible bulk lot
A fraction means part of a bulk lot, not cutting an individual blanket or sealed medicine.

GREEDY LOADING TABLE (highest utility/kg first)
Supply | Ratio | Fraction | Loaded % | Weight loaded kg | Utility gained | Decision
BabyFood | 6.00 | 1.00 | 100.00 | 5.00 | 30.00 | Whole lot
Food | 5.00 | 1.00 | 100.00 | 20.00 | 100.00 | Whole lot
Medicine | 4.00 | 1.00 | 100.00 | 30.00 | 120.00 | Whole lot
Blankets | 3.00 | 0.25 | 25.00 | 5.00 | 15.00 | Partial lot
Water | 2.67 | 0.00 | 0.00 | 0.00 | 0.00 | Not loaded

ITEMS IN THE BOAT
BabyFood: 5.00 kg (100.00% of lot)
Food: 20.00 kg (100.00% of lot)
Medicine: 30.00 kg (100.00% of lot)
Blankets: 5.00 kg (25.00% of lot)
Boat capacity: 60.00 kg
Total weight used: 60.00 kg
Remaining capacity: 0.00 kg
Maximum utility: 265.00
Time: O(N log N) | Storage: O(N)
```